



IIIT Bangalore

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Question 41

Question

Identify the Boolean expression implemented by the given circuit.

Analysis

From circuit analysis the output is

$$Q = A' + B$$

Truth table:

A	B	Q
0	0	1
0	1	1
1	0	0
1	1	1

Python Code

```
import numpy as np
import matplotlib.pyplot as plt

A = np.array([0,0,1,1])
B = np.array([0,1,0,1])

Q = (~A & 1) | B

print("A B Q")

for i in range(4):
    print(A[i], B[i], Q[i])

x = np.arange(4)

plt.step(x,A,where='post',label='A')
```

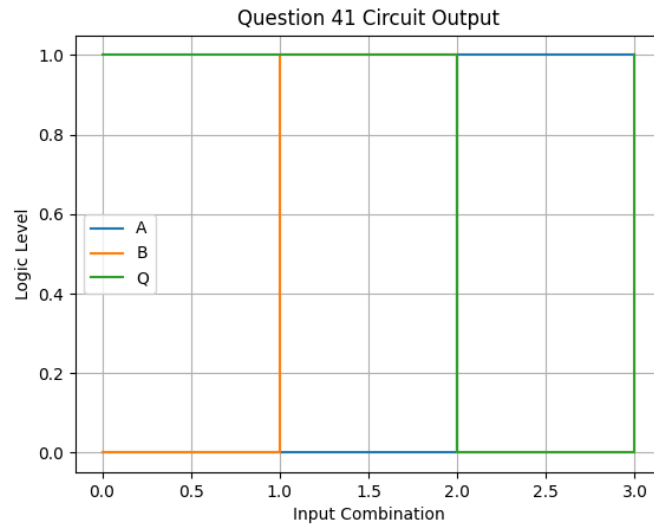
```
plt.step(x,B,where='post',label='B')
plt.step(x,Q,where='post',label='Q')

plt.grid()
plt.legend()

plt.savefig("graph.png")

plt.show()
```

Graph



Conclusion

The output follows the Boolean expression $Q = A' + B$. The graph verifies that the output becomes 1 whenever $A = 0$ or $B = 1$.